CHAPTER III
EXPANSION OF FUNCTIONS

Rolle's Theorem

Statement of Rolle's Theorem: Let a function f be continuous on [*a*,*b*] and differentiable on (*a*,*b*). If $f(a)=0, f(b)=0,$ then there exists at least one point $c \in (a,b)$ such that $f'(c)=0.$

(For Figures, follow the Lecture note)

Rolle's Theorem, in honor of the French Mathematician Michel Rolle (1652-1719), is a special case of Mean-Value Theorem. This theorem states geometrically that if the graph of a differentiable function intersects the x-axis at $x=a$ and at $x=b,$ then there exists at least one point *c* in $c \in (a,b)$ where the tangent line is horizontal.

Example: Find the *x*-intercepts of $f(x)=x^2-5x+4$ and verify that $f'(c)=0$ at some point *c* between those intercepts.

Solution: We have $f(x)=x^2-5x+4=(x-1)(x-4).$ So, the *x*-intercepts of $f(x)$ are $x=1$ and $x=4.$ Since $f(x)$ is a polynomial, it must be continuous and differentiable everywhere. Hence, the hypotheses of Rolle's Theorem are satisfied on [1,4]. But

$$f(a)=f(1)=0, \qquad f(b)=f(4)=0$$

$$f'(x)=2x-5, \qquad f'(c)=2c-5$$

So that $f'(c)=2c-5=0 \Rightarrow c=\frac{5}{2} \in (1,4),$ and consequently, Rolle's Theorem is verified.

Mean-Value Theorem

Statement of Mean-Value Theorem: Let a function f be continuous on [*a*,*b*] and differentiable on (*a*,*b*). Then there exists at least one point $c \in (a,b)$ such that $f'(c)=\frac{f(b)-f(a)}{b-a}.$

(For Figures, follow the Lecture note)

Geometrically, this theorem states that between any two points $A(a, f(a))$ and $B(b, f(b))$ on the graph of a differentiable function *f*, there exists at least one point $c \in (a,b)$ where the tangent line to the graph is parallel to the to the secant line joining points *A* and *B* as shown in Figure 2.

Note that the slope of the secant line *AB* is $\dfrac{f(b)-f(a)}{b-a}$ and so the slope of the tangent line is $f'(c)$. So, $f'(c)=\dfrac{f(b)-f(a)}{b-a}$.

Example: Show that the function $f(x)=\frac{1}{4}x^3+1$ satisfies the hypotheses of the Mean-Value Theorem over the interval [0,2] and find all values of *c* in (0,2) at which the tangent line to the graph of f is parallel to the secant line joining the points $A(a, f(a))$ and $B(b, f(b))$.

Solution: Since $f(x)=\frac{1}{4}x^3+1$ is a polynomial, it must be continuous and differentiable everywhere. In particular, *f* is continuous on [0,2] and differentiable on (0,2). Hence, the hypotheses of the Mean-Value Theorem are satisfied with $a=0$ and $b=2$. But

$$f(a)=f(0)=1, \qquad f(b)=f(2)=3$$

$$f'(x)=\frac{3}{4}x^2, \qquad f'(c)=\frac{3}{4}c^2.$$

So that $f'(c)=\frac{3}{4}c^2=\frac{3-1}{2-0} \Rightarrow c=\pm\frac{2}{\sqrt{3}}$. We see that only $c=\frac{2}{\sqrt{3}}\approx 1.15\in(0,2)$. So, Mean-Value Theorem is verified.

Exercises

1. Verify Rolle's Theorem for the function $f(x)=x^2-3x+2$ over (1,2).
2. Verify Mean-Value Theorem for the function $f(x)=2x-x^2$ over (0,1).

Taylor's theorem is an essential method that provides an approximation of differentiable functions by polynomials. This can be regarded as an extension of the Mean-Value Theorem for higher order derivatives.

We have from the Mean-Value Theorem

$$f(b) - f(a) = (b-a) f'(c) \text{ where } a < c < b$$

$$\Rightarrow \; f(b) = f(a) + (b-a) f'(c) \text{ where } a < c < b \qquad (1)$$

We know that if $a < c < b$, then $c = a + \theta(b-a)$ where $0 < \theta < 1$

Therefore, $f(b) = f(a) + (b-a) f'[a + \theta(b-a)]$ where $0 < \theta < 1$ (2)

Setting $b = x$ in (2), we get

$$f(x) = f(a) + (x-a) f'[a + \theta(x-a)] \text{ where } 0 < \theta < 1 \qquad (3)$$

Setting $b - a = h$ in (2), we get

$$f(a+h) = f(a) + hf'(a + \theta h) \text{ where } 0 < \theta < 1 \qquad (4)$$

Again, setting $a = x$ in (4), we get

$$f(x+h) = f(x) + hf'(x + \theta h) \text{ where } 0 < \theta < 1 \qquad (5)$$

Finally, setting $x = 0$ and $h = x$ in (5), we get

$$f(x) = f(0) + hf'(\theta x) \text{ where } 0 < \theta < 1 \qquad (6)$$

Taylor's Theorem

Statement: If $f(x), f'(x), f''(x), \ldots, f^{(n-1)}(x)$ are continuous on $[a,b]$ and $f^{(n)}(x)$ exists on (a,b), then

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \cdots + \frac{(x-a)^n}{n!} f^{(n)}(a) \qquad (7)$$

Taylor's theorem becomes the Taylor's series when $n \to \infty$.

Taylor's Theorem with Lagrange's form of remainders

If $f(x), f'(x), f''(x), \ldots, f^{(n-1)}(x)$ are continuous on $[a,b]$ and $f^{(n)}(x)$ exists on (a,b), then there exists at least one number $\theta \in (0,1)$ such that

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f^{(3)}(x) + \cdots + \frac{h^n}{n!} f^{(n)}(x + \theta h) \qquad (8)$$

is the Taylor's theorem, where the last term is the Lagrange's form of remainders.

Taylor's Theorem with Cauchy's form of remainders

If $f(x), f'(x), f''(x), \ldots, f^{(n-1)}(x)$ are continuous on $[a,b]$ and $f^{(n)}(x)$ exists on (a,b), then there exists at least one number $\theta \in (0,1)$ such that

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \cdots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(x) + \frac{h^n (1-\theta)^{n-1}}{(n-1)!} f^{(n)}(x+\theta h) \qquad (9)$$

is the Taylor's Theorem, where the last term is the Cauchy's form of remainders.

Maclaurin Theorem

Statement: If $f(x)$ and $f^{(n-1)}(x)$ are continuous on $[a,b]$ and $f^{(n)}(x)$ exists on (a,b), then

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \cdots + \frac{x^n}{n!} f^{(n)}(0) \qquad (10)$$

Maclaurin's theorem becomes the Maclaurin's series when $n \to \infty$.

Maclaurin's Theorem with Lagrange's form of remainders

If $f(x), f'(x), f''(x), \ldots, f^{(n-1)}(x)$ are continuous on $[a,b]$ and $f^{(n)}(x)$ exists on (a,b), then there exists at least one number $\theta \in (0,1)$ such that

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \cdots + \frac{x^{n-1}}{(n-1)!} f^{(n-1)}(0) + \frac{x^n}{n!} f^{(n)}(\theta x) \qquad (11)$$

is the Maclaurin's Theorem, where the last term is the Lagrange's form of remainders.

Maclaurin's Theorem with Cauchy's form of remainders

If $f(x), f'(x), f''(x), \ldots, f^{(n-1)}(x)$ are continuous on $[a,b]$ and $f^{(n)}(x)$ exists on (a,b), then there exists at least one number $\theta \in (0,1)$ such that

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \cdots + \frac{x^{n-1}}{(n-1)!} f^{(n-1)}(0) + \frac{x^n (1-\theta)^{n-1}}{(n-1)!} f^{(n)}(\theta x) \qquad (12)$$

is the Maclaurin's Theorem, where the last term is the Cauchy's form of remainders.

Worked out Problems

1. At what point is the tangent to the curve $y = x^3$ parallel to the chord from points (1, 1) to (2,8)?

Solution: The function $y = f(x) = x^3$ is continuous on [1, 2] and differentiable on (1, 2).

Now $f'(x) = 3x^2$ exists on (1, 2).

By Mean-Value Theorem, we have

$$f(2) - f(1) = (2-1) f'(c) \text{ where } 1 < c < 2$$

$$\Rightarrow \ 8 - 1 = 3c^2 \ \Rightarrow \ c^2 = \frac{7}{3} = 2.33 \ \Rightarrow \ c = 1.15$$

So, $f(1.15) = (1.15)^3 = 2.68$

Hence, the tangent to the given graph is parallel to the chord at (1.15, 2.68).

2. Show that $\ln(x) = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \cdots$ is true over the interval (0, 2].

Solution: We have the Taylor's series

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f^{(3)}(a) + \cdots$$

Take $x = 1 \in (0, 2]$ and let $f(x) = \ln(x)$. Then $f(1) = \ln(1) = 0$.

Also, $f'(x) = 1/x$ and so $f'(1) = 1/1 = 1$; $f''(x) = -1/x^2$ and so $f''(1) = -1$;

$f^{(3)}(x) = 2/x^3$ and so $f^{(3)}(1) = 2$.

Therefore, $f(x) = f(1) + \frac{(x-1)}{1!} f'(1) + \frac{(x-1)^2}{2!} f''(1) + \frac{(x-1)^3}{3!} f^{(3)}(1) + \cdots$

$$= 0 + \frac{(x-1)}{1!}(1) + \frac{(x-1)^2}{2!}(-1) + \frac{(x-1)^3}{3!}(2) + \cdots$$

$$= (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \cdots$$

3. Expand $\ln(1+x)$ in powers of x by Maclaurin series.

Solution: We have the Maclaurin series

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f^{(3)}(0) + \frac{x^4}{4!} f^{(4)}(0) + \cdots$$

Let $f(x) = \ln(1+x)$. Then $f(0) = \ln(1) = 0$.

Also, $f'(x) = 1/(1+x)$ and so $f'(0) = 1/(1+0) = 1$; $f''(x) = -1/(1+x)^2$ and so $f''(0) = -1$;

$f^{(3)}(x) = 2/(1+x)^3$ and so $f^{(3)}(0) = 2$.

Therefore, $f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f^{(3)}(0) + \cdots$

$$= 0 + \frac{x}{1!}(1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(2) + \cdots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

4. Find the Maclaurin series for $\sin x$ in powers of x.

Solution: We have the Maclaurin series

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) - \frac{x^3}{3!} f^{(3)}(0) + \cdots$$

Let $f(x) = \sin x$. Then $f(0) = \sin 0 = 0$.

Also, $f'(x) = \cos x$ and so $f'(0) = \cos 0 = 1$; $f''(x) = -\sin x$ and so $f''(0) = 0$;

$f^{(3)}(x) = -\cos x$ and so $f^{(3)}(0) = -1$; $f^{(4)}(x) = \sin x$ and so $f^{(4)}(0) = 0$.

Therefore, $f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f^{(3)}(0) + \frac{x^4}{4!} f^{(4)}(0) + \cdots$

$$= 0 + \frac{x}{1!}(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(-1) + \frac{x^4}{4!}(-1) + \cdots$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

Exercises

1. Find the first three nonzero terms of the Maclaurin series for the following functions:

(a) $y = e^{-2x}$ (b) $y = \log(\sec x)$

(c) $y = \log(\cos x)$ (d) $y = \sec x$

(e) $y = \cos^2 x$ (f) $y = \tan^{-1}(1+x)$

(g) $y = \log(1+e^x)$ (h) $y = \log(1+\sin x)$

(i) $y = e^x \cos x$

2. Expand $f(x) = \dfrac{1}{x}$ in powers of $(x-2)$.

3. Expand $e^{ax}\sin(px)$ by Maclaurin's theorem with Cauchy's form of remainders.

4. Expand $\log(1+x)$ by Maclaurin's theorem with Lagrange's form of remainders after $2n$ terms.

5. Prove that (i) $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \cdots + (-1)^{n-1}\dfrac{x^{2n-1}}{(2n-1)!} + (-1)^n \dfrac{x^{2n-1}}{(2n)!}\sin(\theta x),\ 0 < \theta < 1$

(ii) $\log(1-x) = -x - \dfrac{x^2}{2} - \dfrac{x^3}{3} - \cdots - \dfrac{x^{n-1}}{n-1} - \dfrac{x^n}{n(1-\theta x)^n},\ 0 < \theta < 1$

6. Expand $\log(1+x)$ by Taylor's theorem with Lagrange's form of remainders.